

# ROGUE · Research Brief

A solo research build of a continuous open-web LLM red-team. The methods and the measured results, including the negative ones.

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## 01 Calibrating an LLM-as-judge against human labels, then recalibrating when a benchmark exposed it.

Every breach verdict is an LLM judgment, so the judge is the load-bearing weakness. It was validated four ways, three against independent human-annotated benchmarks: blind stratified in-distribution hand-labels, WildGuardTest (Allen AI annotators), StrongREJECT, and JailbreakBench's *judge\_comparison* (300 human-labeled rows against four field classifiers). JailbreakBench exposed over-flagging: the v1 judge agreed with the human majority only **70.3%**, **last of five** (behind HarmBench, LlamaGuard-2, GPT-4, Llama-3), at recall 98% and precision 55%.

A 20-row false-positive audit traced it to **five recurring failure modes**. The root cause was a *rubric* problem: it rewarded engagement with the attack frame (persona acceptance, acknowledgment, format mimicry) over transfer of harmful content. A **content-transfer-gate rubric (v3)** moved the same 300 rows to **89.3% agreement, 79.5% precision, 95.5% recall** (a 19-point agreement gain, 24.5-point precision gain, 2.5-point recall cost), lifting it from last to 3rd of five and tied with the frontier classifiers, for about \$8.4 via a tiered evaluation.

Then the honest part. Re-judging the stored breach matrix under v3 cut breach cells from **2,429 to 1,371, a 43.6% reduction**, correcting prior over-reporting. All three external axes were re-measured under v3 (WildGuard harm 88.5%; StrongREJECT 26% more conservative).

|                                            |                                |                                                |                                        |
|--------------------------------------------|--------------------------------|------------------------------------------------|----------------------------------------|
| <b>70.3 → 89.3%</b><br>JBB HUMAN AGREEMENT | <b>55 → 79.5%</b><br>PRECISION | <b>2,429 → 1,371</b><br>BREACH CELLS RE-JUDGED | <b>2.56%</b><br>IN-DIST FALSE-POSITIVE |
|--------------------------------------------|--------------------------------|------------------------------------------------|----------------------------------------|

|                 |                        |             |
|-----------------|------------------------|-------------|
| <b>ROGUE v1</b> | <div><div></div></div> | <b>70.3</b> |
| HarmBench       | <div><div></div></div> | 78.3        |
| LlamaGuard-2    | <div><div></div></div> | 87.7        |
| <b>ROGUE v3</b> | <div><div></div></div> | <b>89.3</b> |
| GPT-4           | <div><div></div></div> | 90.3        |
| Llama-3         | <div><div></div></div> | 90.7        |

Judge agreement vs the field (JailbreakBench, % vs the human majority). Recalibration moved ROGUE from last of five to 3rd, tied with the frontier classifiers.

### WHY THIS IS NOTABLE

A named false-positive taxonomy for a safety judge, plus a measured finding that two respected benchmarks (WildGuardTest harm labels and StrongREJECT) themselves *over-count* relative to a strict content-transfer standard.

## 02 Scheduling as a capability lever, not just an optimization.

A within-tier greedy reorder was replaced with a **target-conditioned cross-tier scheduler**: a static, explainable blend (0.5 global, 0.3 vendor, 0.2 family breach-rate; Laplace-smoothed; deliberately no ML and no bandit, so it stays reproducible). A single-variable controlled experiment (same ladder, attacks, corpus, judge, and target on Claude Haiku across AdvBench and JailbreakBench, with only the order changed) beat the production baseline on every axis: **median winner-rank 22 → 11–13.5**, **attack-success-rate 50% → 60%**, and **cost-per-success \$1.25 → \$0.74 (41% cheaper)**.

|                                      |                                        |                                        |
|--------------------------------------|----------------------------------------|----------------------------------------|
| <b>22 → 11</b><br>MEDIAN WINNER-RANK | <b>50 → 60%</b><br>ATTACK-SUCCESS-RATE | <b>41% cheaper</b><br>COST PER SUCCESS |
|--------------------------------------|----------------------------------------|----------------------------------------|

**WHY THIS IS NOTABLE**

The mechanism is the interesting part: a lower rank *caused* a higher success rate. The old order exhausted the per-scan budget cap before reaching the winning technique, so reordering improved coverage, cost, and latency at once with zero new attacks. The reproducibility invariant is “reorder, never exclude”: same ladder, different order, full reachability preserved.

**03 A publication-grade null result: grammar-component predictive power.**

Before building a grammar/AST attack-composition engine, a **\$0 observational study over 1,540 (primitive × target) cells** tested whether grammar-structure nodes predict breach *beyond* attack-family membership, with full confound controls: Benjamini–Hochberg FDR across hundreds of node and pair tests, Mantel–Haenszel stratification by target model, within-family lift, and Cramér’s-V collinearity flagging. The verdict was **weak to none**: the family label carries the predictive weight, cross-family structural nodes show roughly 1.0 to 1.1× non-significant lift, and the striking pre-FDR pairwise synergies (odds ratios up to 16.8) survived **none** of the four controls.

|                                  |                                              |                                              |
|----------------------------------|----------------------------------------------|----------------------------------------------|
| <b>1,540</b><br>CELLS, \$0 STUDY | <b>~1.0–1.1×</b><br>CROSS-FAMILY LIFT (N.S.) | <b>0 of 4</b><br>SYNERGIES SURVIVED CONTROLS |
|----------------------------------|----------------------------------------------|----------------------------------------------|

**WHY THIS IS NOTABLE**

A cheap, rigorous falsification that redirected engineering away from a months-long build: a successful negative result.

**04 Measure-before-build discipline.**

\$0 measurements from existing telemetry were used repeatedly to *invert* “build it” decisions, each parked with an explicit trigger to revisit:

- **Per-model ladder routing.** The spread was a model main effect, not a family×model interaction, so it was not worth the rewrite.
- **LLM renderer-synthesis.** The synthesis-grade backlog stayed flat at 7 across two widening harvests, so it was parked.
- **HF jailbreak-dataset bulk-import.** It measured 0 new attack families, so it was declined.

**LIMITATIONS, STATED PLAINLY**

Targets are black-box live-API models whose versions are not pinned. Some cells are small-n (95% bootstrap confidence intervals are persisted precisely because of this). The judge is single-operator-calibrated. These are descriptive measurements of a live system, not validated generalizations.